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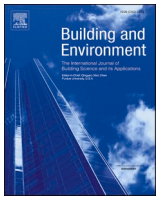

Soundscape diversity governs thermo-physiological and psychological states in a sexually dimorphic manner

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ABSTRACT

Although soundscapes modulate outdoor thermal perception, sex differences in physiological and psychological mechanisms under thermal stress remain elusive. This study examined thermo-acoustic interactions across six acoustic conditions that varied in diversity (low/medium/high) and environmental context (nature-dominant versus anthropogenic-dominant) in male and female participants exposed to graded thermal stress. Physiological indicators (blood pressure, heart rate, and eardrum temperature), electroencephalographic (EEG) activity, and psychological metrics were continuously monitored alongside microclimatic parameters. Results demonstrate marked sexual dimorphism in thermo-acoustic perception, primarily within nature-dominant environments. Males exhibited cardiovascular attenuation under slight thermal stress, with systolic pressure declining 6.08 mmHg (medium diversity) and diastolic pressure 6.61 mmHg (high diversity) versus low diversity ($p < 0.05$), though physiological responsivity diminished as thermal stress escalated. Females conversely demonstrated intensified physiological reactions with increasing thermal load, including reduced heart rate under no thermal stress conditions in anthropogenic settings ($p < 0.05$). In nature-dominant sound environments, low soundscape diversity attenuated male δ power while potentiating α power, consistent with diminished somnolence and heightened relaxation; elevated female θ power, conversely, indexed enhanced positive affect. Psychologically, males demonstrated consistently higher sensitivity, while females exhibited a biphasic pattern of initial attenuation followed by enhancement. In anthropogenic-dominant environments, males displayed heightened physiological and psychological sensitivity to acoustic diversity, whereas females showed asynchronous patterns: physiological responses weakened then strengthened, while psychological responses progressively intensified with increasing thermal stress. Our findings inform sex-sensitive acoustic optimization frameworks for urban design, advancing evidence-based healthy built environment strategies.

1. Introduction

1.1. Background

Human environmental perception is mediated through five sensory modalities, with vision dominantly governing spatial cognition and audition serving as the secondary influential channel [1]. Multisensory integration research demonstrates that perceptual modalities function synergistically rather than independently, converging to construct holistic environmental experiences [2]. The soundscape, defined as the acoustic environment perceived, experienced, and understood within specific contexts, significantly modulates both physiological homeostasis and psychological states [3]. Biophilic acoustic stimuli promote psychophysiological restoration and affective regulation [4–6], whereas anthropogenic noise, particularly traffic-related emissions, elicits annoyance and compromises wellbeing [7,8]. Consequently, outdoor environmental research that privileges visual parameters while neglecting acoustic dimensions remains fundamentally incomplete.

Outdoor thermal perception emerges from the interplay of meteorological parameters and non-thermal environmental modulators, with the acoustic environment serving as a critical regulatory factor [9]. Empirical evidence indicates that biophilic acoustic stimuli mitigate thermal discomfort and enhance environmental acceptability, whereas anthropogenic noise exacerbates thermal stress [10]. Notably, this thermo-acoustic interaction exhibits thermal-intensity dependence:

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Nomenclature

T_a	Air temperature
RH	Relative humidity
V_a	Wind speed
T_g	Global temperature
T_{mrt}	Mean radiant temperature
PET	Physiological equivalent temperature
TSV	Thermal sensation vote
MTSV	Mean thermal sensation vote
TCV	Thermal comfort vote
SDV	Soundscape diversity sensation vote
SPV	Soundscape diversity pleasantness vote
OS	Open space
PS	Partially shaded space
SS	Shaded space
SVF	Sky view factor
ROS	Restoration Outcome Scale
PANAS	Positive and Negative Affect Schedule
EEG	Electroencephalogram
PSD	Power spectral density
ICA	Independent component analysis
BP	Blood pressure
SBP	Systolic blood pressure
DBP	Diastolic blood pressure
HR	Heart rate
ET	Eardrum temperature
SDI	Soundscape diversity index
SRI	Soundscape richness index
SDO	Soundscape dominance index
SEI	Soundscape evenness index
ND	Nature-dominant
AD	Anthropogenic-dominant
LSD	Low soundscape diversity
MSD	Medium soundscape diversity
HSD	High soundscape diversity
PLS-VIP	Partial least square-variable importance in projection
VIP	Variable Importance in Projection
SCS	Slight cold stress
NTS	No thermal stress
STS	Slight thermal stress
MTS	Moderate thermal stress

soundscape benefits are amplified under heat stress but attenuated during cold or neutral exposure [11].

Soundscape-mediated thermal mitigation has emerged as a pivotal research frontier in landscape architecture and built environment studies. As a perceptual regulatory strategy, acoustic optimization offers a cost-effective alternative to energy-intensive physical cooling interventions [12]. Systematic elucidation of these thermo-acoustic mechanisms and their translation into spatial design strategies thus hold substantial implications for advancing urban environmental quality.

1.2. Literature review

Soundscapes modulate autonomic function beyond subjective perception, operating through central and peripheral pathways to influence cardiovascular and electrodermal activity, as well as affective dimensions of valence and arousal [13]. These psychophysiological effects are source-dependent: natural stimuli generally facilitate stress recovery, whereas anthropogenic noise elicits adverse autonomic and emotional responses [14–16]. Sound pressure level represents a critical physical determinant of these responses [17,18].

As soundscape assessment has shifted from unidimensional metrics to multidimensional structural complexity, soundscape diversity has emerged as a key indicator of environmental information richness [19]. Particularly within urban green spaces, diversity significantly mediates acoustic preference, affective quality, and overall soundscape appraisal, serving as a critical bridge between spatial characteristics and perceptual evaluation [20,21]. Elevated biophonic diversity promotes psychophysiological restoration [22]; however, this relationship follows an inverted-U pattern, wherein excessive acoustic layering attenuates beneficial effects [23].

Thermal environments modulate cognitive and affective states, thereby governing perceptual processing of ambient stimuli [24]. Thermal stress activates autonomic thermoregulatory mechanisms, manifesting as altered cardiovascular activity, cutaneous temperature, and endocrine profiles [25]. In actual outdoor contexts, thermal and acoustic parameters exert synergistic effects on environmental perception and comfort evaluation [12]. Elevated sound pressure levels amplify thermal sensation and compromise thermal comfort, whereas biophilic soundscapes attenuate perceived warmth relative to anthropogenic acoustic environments [26,27]. This thermo-acoustic interaction is stress-dependent: under moderate thermal loads, acoustically pleasant environments significantly diminish subjective warmth sensation [24], while acoustic dissatisfaction predicts heightened heat discomfort [28]. Reciprocally, microclimatic parameters, encompassing air temperature, humidity, and solar radiation, critically shape soundscape appraisal in open-space settings [29,30].

Sex constitutes a critical individual difference variable mediating psychophysiological reactivity to acoustic environments [31,32]. Females demonstrate heightened auditory sensitivity and noise susceptibility, whereas males exhibit exaggerated cardiovascular and autonomic responses to acoustic stimulation [33]. Sexual dimorphism extends to soundscape preferences; for example, females demonstrate particular affinity for wind sounds [34].

These disparities persist under thermal challenge. Females display enhanced thermal sensitivity and more volatile subjective thermal evaluations than males under identical microclimatic conditions [35], potentially reflecting fundamental differences in adipose distribution and metabolic thermoregulation [36]. Consequently, females exhibit elevated vulnerability to thermal fatigue and negative affective states. Thus, soundscape diversity effects likely manifest sexually dimorphic patterns across thermal stress gradients, necessitating systematic investigation.

1.3. Research gap

Extant research on soundscape-thermo-physiological interactions has predominantly examined acoustic intensity and source typologies, yet remains largely unidimensional, neglecting structural complexity and diversity effects. Critically, the mechanisms by which soundscape diversity modulates psychophysiological responses across thermal stress gradients remain elusive. Furthermore, the moderating effects of sex on these restorative effects has been largely overlooked, with sexual dimorphism in diversity perception and thermo-physiological reactivity poorly characterized. This necessitates a systematic, sex-differentiated investigation of soundscape diversity mechanisms to inform nuanced, sex-sensitive design interventions for urban open spaces, ultimately advancing population health and wellbeing.

1.4. Research objectives

This study was conducted across three typologically distinct urban

spaces in Xi'an (cold-climate region, China), characterized by gradient shading intensities. A multimodal assessment protocol was deployed, integrating microclimatic characterization, psychometric evaluation, cardiovascular and thermal physiology monitoring, and electroencephalographic (EEG) recording. Six acoustic conditions, encompassing low (LSD), medium (MSD), and high (HSD) diversity within both nature-dominant and anthropogenic-dominant contexts, were experimentally manipulated. Sex-differentiated perceptual, physiological, and psychological responses were systematically characterized, and the relative contributions of acoustic diversity versus microclimatic parameters were quantified. Specifically, this research sought to: (1) elucidate sexual dimorphism in perceptual responses to acoustic diversity across thermal stress gradients; (2) characterize physiological and psychological correlates of soundscape diversity exposure; and (3) ascertain the relative influence weights of acoustic versus microclimatic determinants on thermo-acoustic perception and psychophysiological outcomes. These findings advance theoretical understanding of sex-specific thermo-acoustic mechanisms, informing evidence-based, sex-sensitive soundscape design strategies for urban thermal mitigation and sustainable built environment development.

2. Method

2.1. Experimental design

2.1.1. Study site

Xi'an occupies the Köppen climate boundary between cold semi-arid (BSk) and humid subtropical (Cwa) regimes, exhibiting a semi-humid continental monsoon climate with pronounced seasonal contrast, characterized by hot, rainy summers and cold, dry winters [37]. Local climatological records indicate maximum monthly mean air temperature (T_a) of 28.08 °C in July (absolute maximum 39.33 °C), and minimum monthly mean T_a of 0.62 °C in January (absolute minimum −8.01 °C), with annual relative humidity (*RH*) ranging 51.28–73.68 % [38].

The trials were conducted at a 170 ha university campus in Xi'an (34°15′44″N, 108°04′37″E), selected for its morphological heterogeneity. Three microclimatic contexts, including open (OS), partially shaded (PS), and shaded (SS), were stratified by landscape configuration and Sky View Factor (SVF), with site selection prioritizing acoustic isolation to preclude ambient noise interference (Fig. 1). SVF was quantified via hemispherical photography (Canon EOS 6D; EF 8–15 mm f/4 L fisheye lens) and subsequent RayMan analysis.

2.1.2. Soundscape selection

Soundscape diversity and source typology significantly modulate emotional appraisal and overall acoustic evaluation [20,24,39,40]. Accordingly, six experimental conditions were synthesized via orthogonal combination of three diversity levels (LSD, MSD, and HSD) and two dominant source contexts: natural versus anthropogenic. Ten elemental sources were curated: five natural (birdsong, flowing water, insect chirps, wind, and frog croaks) and five anthropogenic (traffic, human conversation, children's play, horn sounds, and construction noise), obtained from open-access acoustic repositories (https://www.freesfx.co.uk/soundeffects; https://freesound.org). Using Adobe Audition 2024, these were mixed to achieve distinct Soundscape Diversity Index (SDI), Richness Index (SRI), Dominance Index (SDO), and Evenness Index (SEI) indices while maintaining consistent source typology per condition. Each 15-minute stimulus was normalized for sound pressure

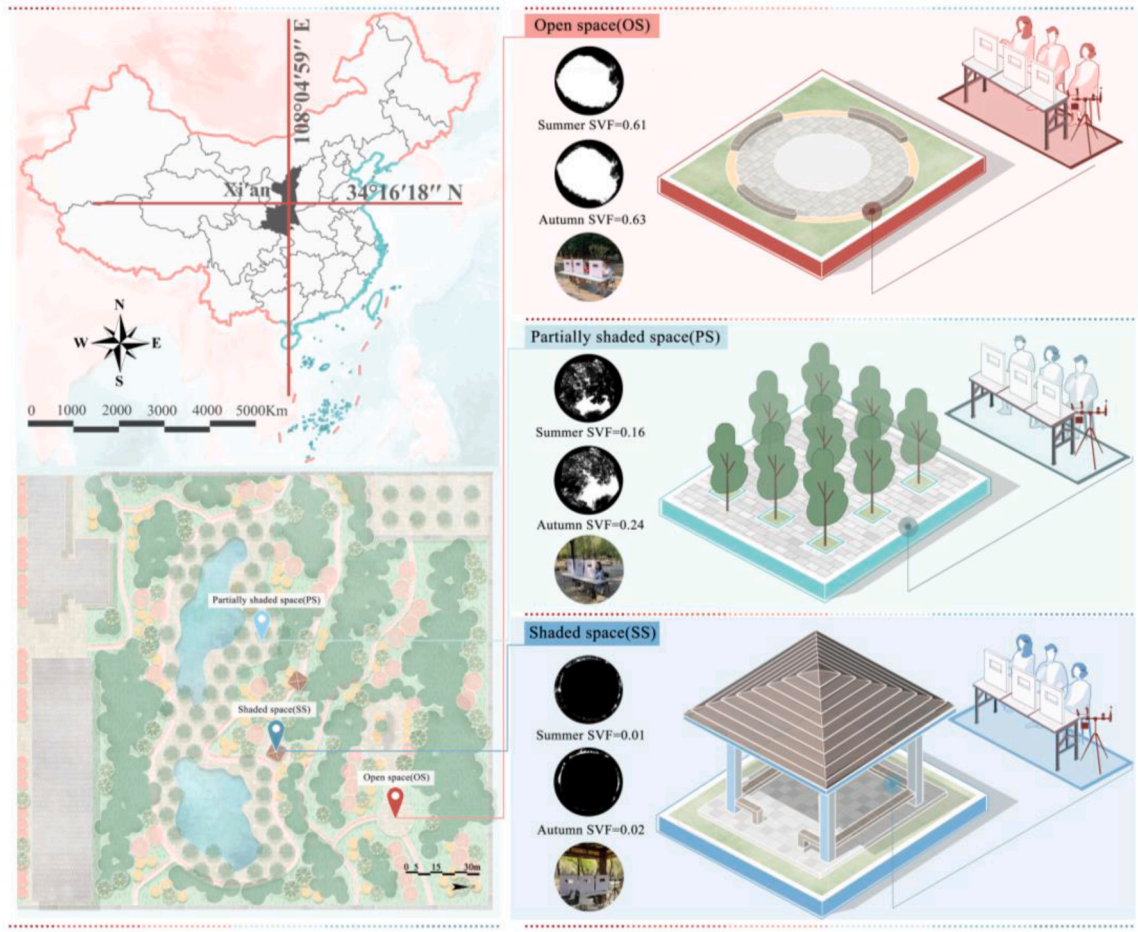

Fig. 1. The study area and measured sites.

level to ensure perceptual clarity without auditory masking (Table 1, Fig. 2). Prior research demonstrates that perceived acoustic comfort is independent of sound pressure level within the 65–70 dBA range [2,41]. Accordingly, playback levels were calibrated to 65–70 dBA using an AWA5688 multifunction sound level meter. The campus environment, designated for teaching and learning, inherently maintains low ambient noise with negligible traffic intrusion, ensuring comparable background sound pressure levels across experimental sites. To control for extraneous acoustic interference, trials were temporally isolated from campus events, and pedestrian access near the experimental areas was restricted by research personnel during data collection.

2.1.3. Participants selection

A total of 864 right-handed participants (Male:Female = 1:1.2; *Appendix Table A*) were recruited under stringent inclusion criteria: normal auditory and cognitive function; non-native English proficiency sufficient for psychometric assessment; capacity for independent procedural comprehension and questionnaire completion; stable metabolic status (adequate sleep, balanced diet); 3 h pre-testing abstention from psychoactive substances (nicotine, caffeine, alcohol); and clean scalp for optimal EEG signal acquisition.

2.1.4. Experimental procedure

Experimental data were acquired during two seasonal campaigns (30 July–4 August and 2–8 November 2025), encompassing 12 clear-sky days with eight daily sessions. Consistent sunny and calm conditions within each season standardized thermal parameters. Each session randomized nine participants across three microclimatic contexts to one of six acoustic conditions; participants underwent single-condition exposure per session (*Appendix Table B*). Conditions were counterbalanced via a modified Latin-square design and evenly distributed across contexts [42]. Prior to each session, loudspeakers were concealed from view and preloaded with acoustic stimuli. Loudspeaker placement, participant distance, exposure duration, and procedural parameters were standardized across participants. The protocol comprised three standardized phases:

Stage 1: Acclimatization (15 min). Following informed consent and EEG electrode placement, participants underwent seated rest with standardized visual fixation (viewfinder-controlled landscape element ratios) to stabilize autonomic tone. Five-minute basal electroencephalographic recording preceded demographic data collection and triplicate autonomic assessment (blood pressure [BP], heart rate [HR], ear temperature [ET]), with mean values derived for baseline metrics.

Stage 2: Stress Induction (5 min). Participants completed an English-language stressor task, an established psychophysiological stressor, with continuous EEG monitoring. Post-task psychometric evaluation and triplicate autonomic measurements followed [43].

Stage 3: Acoustic Restoration (15 min). Randomized exposure to one of six soundscape conditions was administered with concurrent EEG recording. The protocol concluded with post-intervention psychometric assessment and final triplicate autonomic measurements (Fig. 3).

Table 1
The formula and implications of soundscape diversity indices.

Name	Formula	Implication
SDI	$SDI = -\sum_{i=1}^{S} P_i(\ln P_i)$	Reflects the number of sound source species and the uniformity of individual distribution in the species.
SRI	$SRI = S$	Reflects the number of types of sounds.
SEI	$SEI = \mathrm{SDI}/\ln S$	Reflects the evenness of the distribution of individual numbers of sound source types.
SDO	$SDO = \sum_{i=1}^{S} P_i^2$	Reflects the dominance of sound sources in the site.

Notes: Pi is the ratio of the i th sound source individual number to the total number of all sound source individuals in the site; S is the number of types of sound sources in the site.

2.2. Measurements

2.2.1. Meteorological measurements

During the experiment, T_a, RH, wind speed (V_a), and global temperature (T_g) were measured on site. T_a and RH were recorded using a HOBO Onset U23–001 recorder (ONSET, USA), while V_a and T_g were measured using a Kestrel 5500 (Nielsen-Kellerman Co., USA) and a Delta OHM HD2017.2 (Delta OHM, Italy), respectively (*Appendix Table C; Appendix Table D*). All instruments were set to record at 1-min intervals and were positioned at a height of 1.1 m above ground level. Mean radiant temperature (T_{mrt}) was calculated according to ISO Standard 7726 (ISO, 1998):

$$T_{mrt} = \left[(T_g + 273)^4 + \frac{1.10 \times 10^8 V_a^{0.6}}{\varepsilon D^{0.4}} (T_g - T_a) \right]^{\frac{1}{4}} - 273 \quad (1)$$

where ε is the reflectivity of the black globe ($\varepsilon = 0.95$); D is the diameter of the black globe ($D = 0.05$ m).

2.2.2. Questionnaire survey

The questionnaire comprised three sections: demographic information, subjective perception votes, and emotion scales (*Appendix Fig. A*). Demographic data encompassed age, sex, stature, body mass, and clothing insulation [44], collected once during thermal acclimatization with participants remaining seated. Perceptual assessment included Thermal Sensation Vote (TSV; 7-point ASHRAE Standard 55 scale; ASHRAE, 2020), Thermal Comfort Vote (TCV; 5-point symmetric scale), Soundscape Diversity Vote (SDV; 7-point richness scale), and Soundscape Diversity Pleasantness Vote (SPV; 7-point hedonic scale).

Affective states were evaluated using the Restoration Outcome Scale (ROS) and Positive and Negative Affect Schedule (PANAS). ROS, validated for environmental restoration research [45,46], comprises six items assessing stress regulation, attentional recovery, and cognitive clarity (7-point Likert scale; 1 = strongly disagree, 7 = strongly agree), with mean scores representing restorative experience magnitude. PANAS quantified orthogonal positive and negative affect via twenty adjectives (ten per dimension) on 5-point scales (1 = very slightly/not at all, 5 = extremely) [47,48]. Operational clarifications were provided for ambiguous items to ensure valid construct measurement.

2.2.3. Physiological measurements

Cardiovascular (systolic/diastolic blood pressure [SBP/DBP], heart rate [HR]) and thermal (eardrum temperature [ET]) metrics were quantified via automated oscillometry (Omron HEM-7211, Japan) and infrared tympanic thermometry (Braun IRT6520), respectively. Blood pressure assessment followed standardized seated protocols: arm supported at heart level with cuff positioned 1–2 cm above the antecubital fossa [49]. Tympanic temperature was measured with probe orientation toward the nasal bridge to ensure alignment with the tympanic membrane [50]. Triplicate measurements were obtained per condition, with mean values retained for statistical analysis.

2.2.4. Psychological measurements

Electroencephalography (EEG) provides validated assessment of psychophysiological states [51–53]. Spectral power was quantified across four functionally defined frequency bands: δ (1–4 Hz; somnolence/fatigue), θ (4–8 Hz; affective processing), α (8–14 Hz; relaxation), and β (14–30 Hz; attention/arousal) [52,54]. Neural signals were acquired using a 14-channel EPOC X wireless headset (EMOTIV Inc., USA; AF3, AF4, F3, F4, F7, F8, FC5, FC6, T7, T8, P7, P8, O1, O2) with bilateral references (CMS, DRL), arranged according to the International 10–20 system [55]. Data were sampled at 2048 Hz, downsampled to 128 Hz (7.8-ms resolution), and wirelessly transmitted to a tablet interface (Fig. 4). Throughout recording, participants maintained postural immobility and abstained from verbalization and electronic device usage to minimize motion and electromagnetic artifacts. Sensor felts

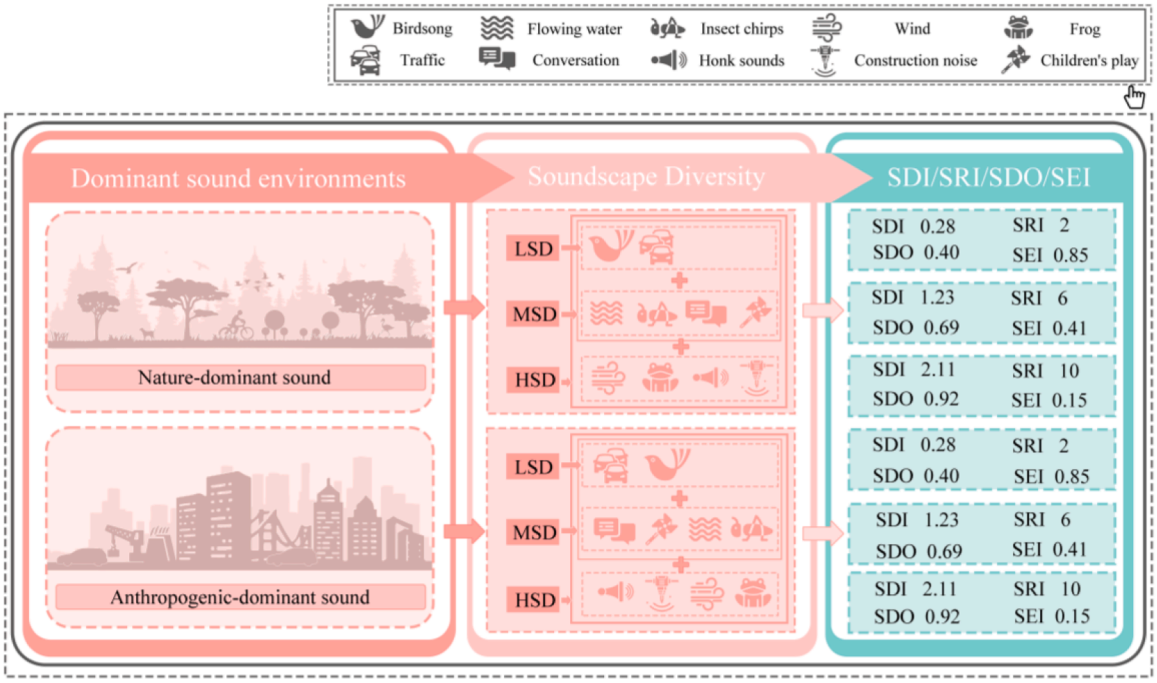

Fig. 2. Soundscape diversity.

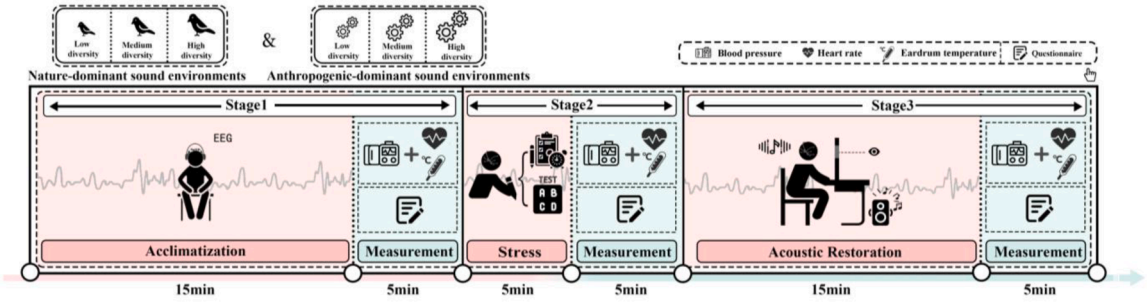

Fig. 3. Experimental procedure.

were hydrated with 0.9 % saline to optimize conductivity. Neural data were acquired via EmotivPRO with real-time quality monitoring and exported to MATLAB for spectral decomposition.

2.3. Thermal index

Mean thermal sensation votes (MTSV) were computed across experimental phases (baseline, stress, restoration) within 1 °C Physiological Equivalent Temperature (PET) strata to parameterize thermal comfort functions [52]. Thermal stress categories were operationalized via PET thresholds: slight cold stress (SCS; $-1.5 \leq$ TSV < -0.5; PET 0.54–15.82 °C), no thermal stress (NTS; $-0.5 \leq$ TSV < 0.5; PET 15.82–31.09 °C), slight thermal stress (STS; $0.5 \leq$ TSV < 1.5; PET 31.09–46.36 °C), and moderate thermal stress (MTS; $1.5 \leq$ TSV < 2.5; PET 46.36–61.63 °C) (Fig. 5).

2.4. Statistical analysis

Data preprocessing entailed outlier detection and missing value treatment. Differential effects of soundscape diversity across thermal stress gradients and dominant source contexts were analyzed via parametric (one-way ANOVA) or nonparametric (Kruskal–Wallis) procedures, contingent upon distributional properties (SPSS v26.0).

2.4.1. EEG

EEG preprocessing was conducted in MATLAB using EEGLAB v14.0.0 [56]. Artifacts were attenuated via visual inspection, band-pass filtration (1–30 Hz, encompassing δ, θ, α, β bands), and independent component analysis (ICA) to remove ocular and myogenic activity. Cardiac artifacts were automatically detected and rejected using the ADJUST algorithm. Power spectral density (PSD) was subsequently derived via fast Fourier transform (FFT) [57].

$$\mathrm{F[k]} = \sum_{n=0}^{N-1} x[n] e^{-\frac{j2\pi kn}{N}} k = 0, \ldots, N-1 \tag{2}$$

$$P(k) = \lim_{n\to\infty} \left(\frac{|F(k)|}{N} \right)^2 \tag{3}$$

where F is the Fourier transform, x is the signal, and P is the PSD.

Power spectral analyses employed normalized (relative) power estimates to account for inter-subject variability [58]. Relative power values were derived for the δ, θ, α, and β bands across all 14 channels as follows:

$$P_{nor} = \frac{P_i}{\sum_{i\in\ (\delta,\theta,\alpha,\beta)} P_i} \tag{4}$$

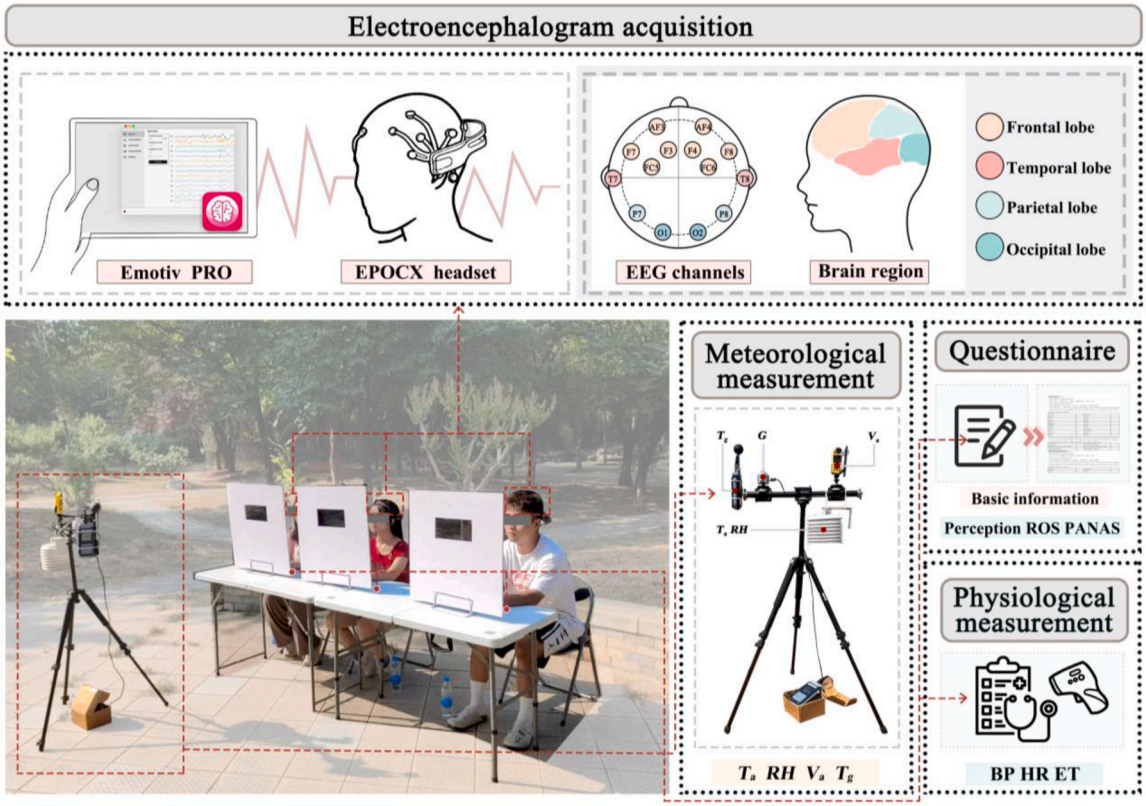

Fig. 4. Schematic of the trial setup.

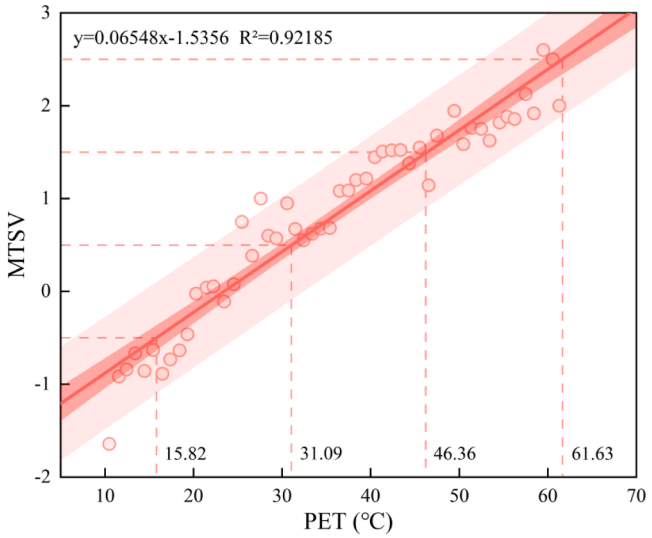

Fig. 5. Correlation between PET and MTSV.

where P_i represents the absolute power of each frequency band, and the normalized power spectrum is the ratio of each frequency band to the sum of the absolute power of four frequency bands (δ, θ, α, and β).

2.4.2. Partial least square-variable importance in projection

Partial least squares–variable importance in projection (PLS-VIP) analysis quantified predictor significance across thermal stress gradients [59]. Within the PLS regression framework, standardized predictor (soundscape diversity and meteorological parameters) and response (perceptual, physiological, and psychological indicators) matrices were fitted to characterize variable contributions to the multivariate system. Predictor and response variables were standardized to eliminate scale effects prior to model fitting [60]. VIP scores quantify each predictor's relative contribution to PLS components and explained response variance [61]. With squared values scaled to unit mean across predictors, scores exceeding 1.0 indicate above-average explanatory power and were retained as influential predictors [62,63].

$$VIP_j = \sqrt{\frac{p\sum_{a=1}^{A}\left[q_a^2\cdot\left(w_{ja}/\|\,w_a\,\|\right)^2\right]}{\sum_{a=1}^{A}q_a^2}} \tag{5}$$

where VIP_j is the VIP value of the j-th variable; p is the total number of predictor variables; A represents the total number of PLS components; q_a stands for the weight of the a-th component; w_{ja} is the weight of the j-th variable in the a-th PLS component; $\|\,w_a\|$ is the Euclidean norm of the weight vector for the a-th PLS component.

3. Results

3.1. Thermal and auditory perceptions

3.1.1. TSV and TCV

Kruskal–Wallis analyses revealed sexual dimorphism in thermo-acoustic perception contingent upon dominant source context. Within natural environments, soundscape diversity did not modulate male TSV ($p > 0.05$) but significantly predicted TCV under STS conditions ($p < 0.05$). Conversely, under slight cold stress, diversity significantly predicted female TSV ($p < 0.05$) without affecting TCV ($p > 0.05$). In anthropogenic- settings, diversity exerted no significant effects on either TSV or TCV for either sex ($p > 0.05$). These results demonstrate that soundscape diversity effects on thermal perception are sexually dimorphic and context-dependent, with pronounced sex differences emerging primarily within nature-dominant acoustic environments (*Appendix Table E*).

Mann–Whitney U analyses revealed sexually antagonistic thermal perceptual responses to acoustic diversity within nature-dominant environments. Under SCS conditions, diversity minimally affected male TSV, whereas HSD significantly elevated female TSV relative to LSD ($p < 0.05$), suggesting acoustic enrichment mitigates subjective cold sensation in females.

Under NTS conditions, MSD and HSD elevated male TSV above LSD levels, while producing the inverse pattern in females. During MTS, MSD diminished male TSV compared with LSD and HSD, yet elevated female TSV. These results demonstrate consistent sexual dimorphism in thermo-acoustic perception, wherein acoustic diversity elicits opposing thermal perceptual effects across sexes (Fig. 6a, b).

Mann–Whitney U analyses further revealed sexually dimorphic effects of acoustic diversity on thermal comfort within nature-dominant environments. Under NTS conditions, diversity minimally modulated male TCV, whereas MSD significantly elevated female thermal comfort above LSD and HSD levels, indicating heightened female responsivity to acoustic complexity. Under STS conditions, LSD significantly improved male TCV relative to MSD ($p < 0.05$), while exerting negligible effects on females, demonstrating reversed sexual dimorphism in thermo-acoustic comfort responsivity. During MTS conditions, HSD aggravated male thermal discomfort but alleviated female discomfort, revealing sexually antagonistic patterns wherein acoustic enrichment elicits opposite comfort trajectories across sexes (Fig. 6c, d).

3.1.2. SDV and SPV

Kruskal–Wallis analyses revealed sexually dimorphic perceptual responses to acoustic diversity contingent upon dominant source context. Within natural environments, diversity significantly modulated male SDV (SCS and MTS) and SPV (SCS) ($p < 0.05$), without affecting female perception ($p > 0.05$). Conversely, in anthropogenic-dominant settings, diversity significantly influenced SDV in both sexes under SCS, and SPV under NTS conditions ($p < 0.05$), with females exhibiting additional significant SDV responsivity ($p < 0.001$). These results indicate that acoustic diversity effects on soundscape perception are sexually dimorphic and context-specific, with pronounced sex differences emerging primarily within nature-dominant environments (*Appendix Table F*).

Mann–Whitney U analyses revealed sexually dimorphic discriminatory capacity for acoustic diversity within natural environments. Under SCS conditions, males exhibited significantly attenuated SDV under LSD relative to HSD ($p < 0.05$), whereas female responsivity was negligible, indicating heightened male perceptual acuity for complexity. During STS, equivalent discrimination emerged in both sexes. Conversely, under MTS conditions, sexually antagonistic patterns emerged: males discriminated HSD, whereas females discriminated LSD, suggesting sex-specific perceptual anchoring to diversity extremes.

Within anthropogenic-dominant environments, sexual dimorphism in acoustic diversity discrimination varied with thermal stress intensity. Under SCS conditions, both sexes discriminated LSD from higher diversity levels, though males exhibited larger effect sizes than females ($p < 0.05$). During NTS, male responsivity was attenuated, whereas females discriminated LSD from MSD and HSD with high significance ($p < 0.001$). This pattern persisted under STS: males showed negligible discrimination, while females reliably distinguished LSD from MSD/HSD. Conversely, under MTS conditions, sexually antagonistic patterns emerged: males discriminated LSD, whereas females discriminated HSD, indicating sex-specific perceptual salience of complexity extremes (Fig. 7a, b).

Within anthropogenic-dominant environments, acoustic diversity elicited sexually dimorphic hedonic responses across thermal gradients. Under SCS conditions, LSD significantly diminished male pleasantness relative to MSD ($p < 0.05$), whereas female SPV exhibited minimal diversity-dependent modulation. During NTS, sexually antagonistic patterns emerged: HSD attenuated male pleasantness, while LSD attenuated female pleasantness, indicating divergent acoustic preferences. Conversely, under MTS conditions, HSD enhanced male pleasantness, whereas MSD enhanced female pleasantness, revealing sex-specific optima for acoustic complexity (Fig. 7c, d).

3.2. Physiological responses

3.2.1. Systolic blood pressure

One-way ANOVA revealed sexually dimorphic cardiovascular responsivity to acoustic diversity. The SBP was significantly modulated in males under STS conditions within natural environments ($p < 0.05$), whereas females exhibited no significant modulation ($p > 0.05$), indicating context-dependent male susceptibility. The DBP showed similar sexual dimorphism, with significant male responses under SCS (natural) and NTS (anthropogenic) ($p < 0.05$), contrasting with female non-responsivity ($p > 0.05$). These findings demonstrate that male cardiovascular regulation is more sensitive to soundscape diversity than female responses across varying thermal and acoustic contexts (*Appendix Table G*).

Specifically, within natural environments, males demonstrated a quadratic SBP response to diversity under NTS and MTS conditions, whereas females exhibited converse U-shaped trajectories. Critically, under STS conditions, MSD significantly attenuated male SBP relative to LSD (−6.08 mmHg, $p < 0.05$), suggesting optimal cardiovascular regulation at intermediate complexity levels (Fig. 8a, b).

In natural contexts, male DBP declined linearly with increasing diversity under STS but exhibited biphasic trajectories under NTS and MTS conditions, whereas female responses were more heterogeneous. Critically, HSD significantly attenuated male DBP relative to LSD under STS conditions (−6.61 mmHg, $p < 0.05$), indicating maximal cardiovascular benefit at highest acoustic complexity (Fig. 8c, d).

3.2.2. Heart rate and eardrum temperature

One-way ANOVA revealed sexually dimorphic chronotropic responsivity: male HR was significantly modulated under STS within natural environments ($p < 0.05$), whereas female HR exhibited significant responsivity under NTS anthropogenic conditions ($p < 0.05$). Conversely, ET showed no significant modulation across conditions ($p > 0.05$), indicating negligible thermoregulatory influence of acoustic diversity (*Appendix Table H*).

Within natural environments under STS conditions, LSD significantly attenuated male HR relative to MSD ($p < 0.05$). Conversely, under NTS anthropogenic conditions, HSD significantly lowered female HR compared to LSD ($p < 0.05$) (Fig. 9a, b). Under SCS and NTS conditions, males exhibited minimal tympanic temperature at MSD, whereas females showed minimal temperatures at HSD. Under STS and MTS conditions, male ET minima shifted to LSD, while females maintained HSD optima. Within anthropogenic environments, males consistently demonstrated MSD-associated ET minima across thermal gradients, whereas females exhibited no clear diversity-dependent thermoregulatory pattern (Fig. 9c and d).

3.3. Psychological effects

3.3.1. ROS

Within natural environments, sexual dimorphism characterized restorative outcomes. For males, MSD consistently facilitated stress recovery across thermal gradients, whereas attentional and cognitive benefits were thermally contingent: HSD maximally enhanced attentional recovery under SCS conditions, while MSD optimally promoted cognitive clarity. In contrast, females demonstrated consistent stress recovery and cognitive benefits under LSD across all thermal conditions. Notably, HSD's cognitive efficacy diminished with escalating thermal stress, while MSD exhibited minimal restorative effects. Attentional recovery in females was thermally contingent except under STS conditions, with HSD proving most effective under NTS conditions (Table 2).

Within anthropogenic environments, HSD consistently exerted

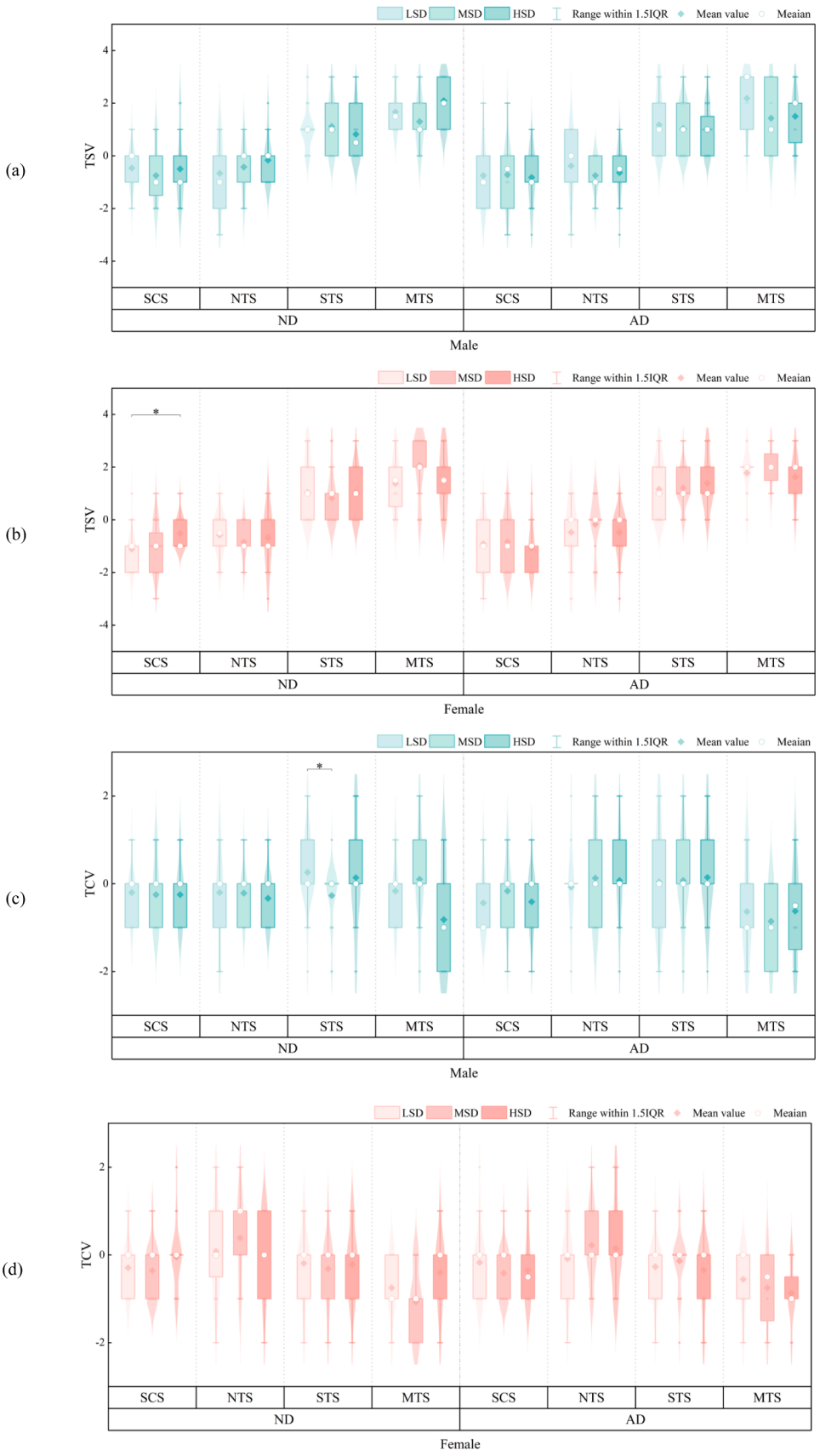

Fig. 6. Thermal perceptions of male and female in different thermal stress and soundscape diversity across different sound source types: (a) male TSV, (b) female TSV, (c) male TCV, (d) female TCV.

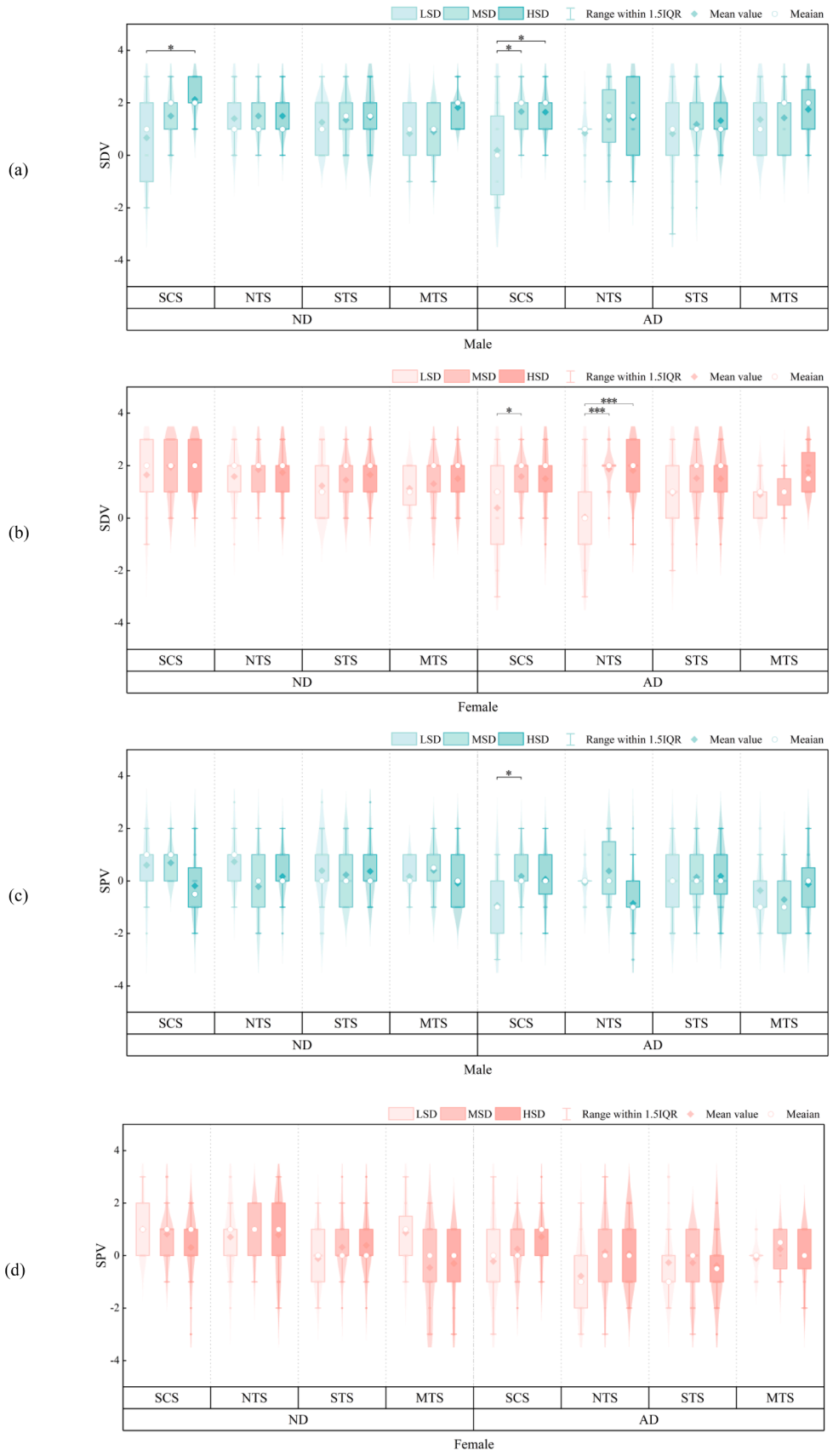

Fig. 7. Auditory perceptions of male and female in different thermal stress and soundscape diversity across different sound source types: (a) male SDV, (b) female SDV, (c) male SPV, (d) female SPV.

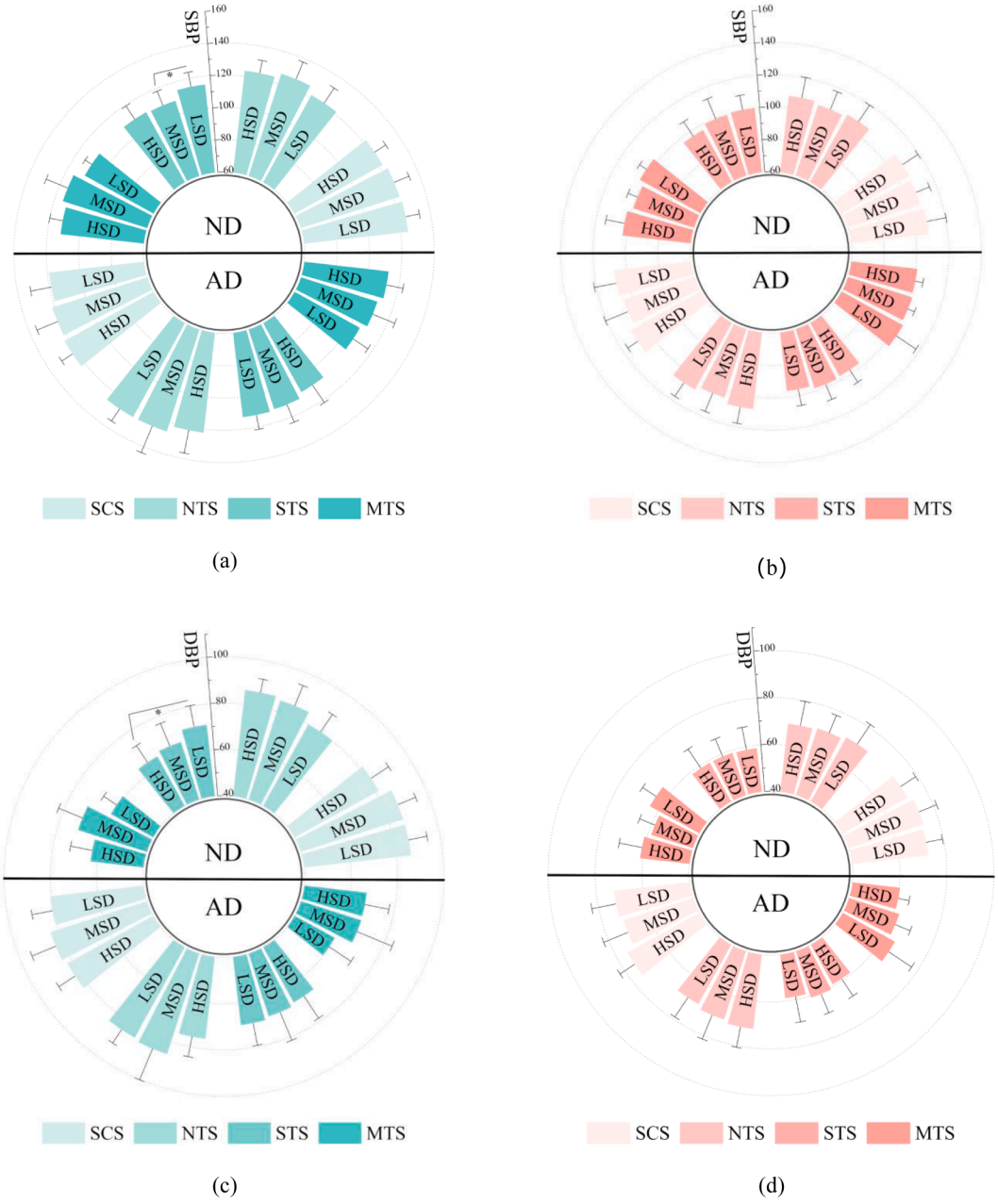

Fig. 8. Systolic blood pressure and diastolic blood pressure of male and female in different thermal stress and soundscape diversity across different sound source types: (a) male SBP, (b) female SBP, (c) male DBP, (d) female DBP.

beneficial effects on male stress recovery and cognitive function across thermal gradients. LSD demonstrated increasing restorative efficacy with escalating thermal stress, whereas MSD exhibited converse patterns. Attentional outcomes were thermally contingent, with MSD proving optimal under NST conditions. Conversely, females exhibited stress recovery benefits under both MSD and HSD across thermal conditions. HSD-mediated attentional benefits intensified with thermal stress, while cognitive outcomes were thermally stratified: HSD proved maximally effective under NTS conditions, whereas alternative diversity levels predominated under thermal challenge (Table 3).

3.3.2. PANAS

Within natural environments, affective responses to acoustic diversity exhibited marked sexual dimorphism across thermal gradients. Under SCS conditions, males demonstrated optimal emotional restoration at MSD, whereas females achieved optimal affective states at LSD. During NTS, MSD maximally enhanced male emotional valence. Under STS conditions, HSD optimized male affective recovery; conversely, females exhibited divergent optima: LSD maximized PA while HSD minimized NA. Under MTS conditions, LSD provided insufficient emotional restoration for males, whereas females achieved complete affective recovery under LSD (Fig. 10a).

Within anthropogenic-dominant contexts, affective restoration exhibited marked sexual dimorphism. Under NTS conditions, males achieved complete affective optimization under MSD, whereas females failed to restore pre-stress baselines despite MSD-induced valence shifts. Under STS conditions, HSD maximally improved male affective states; conversely, females exhibited only partial affective modulation under LSD and HSD. During MTS, LSD facilitated complete male affective restoration, while MSD produced incomplete female recovery (Fig. 10b).

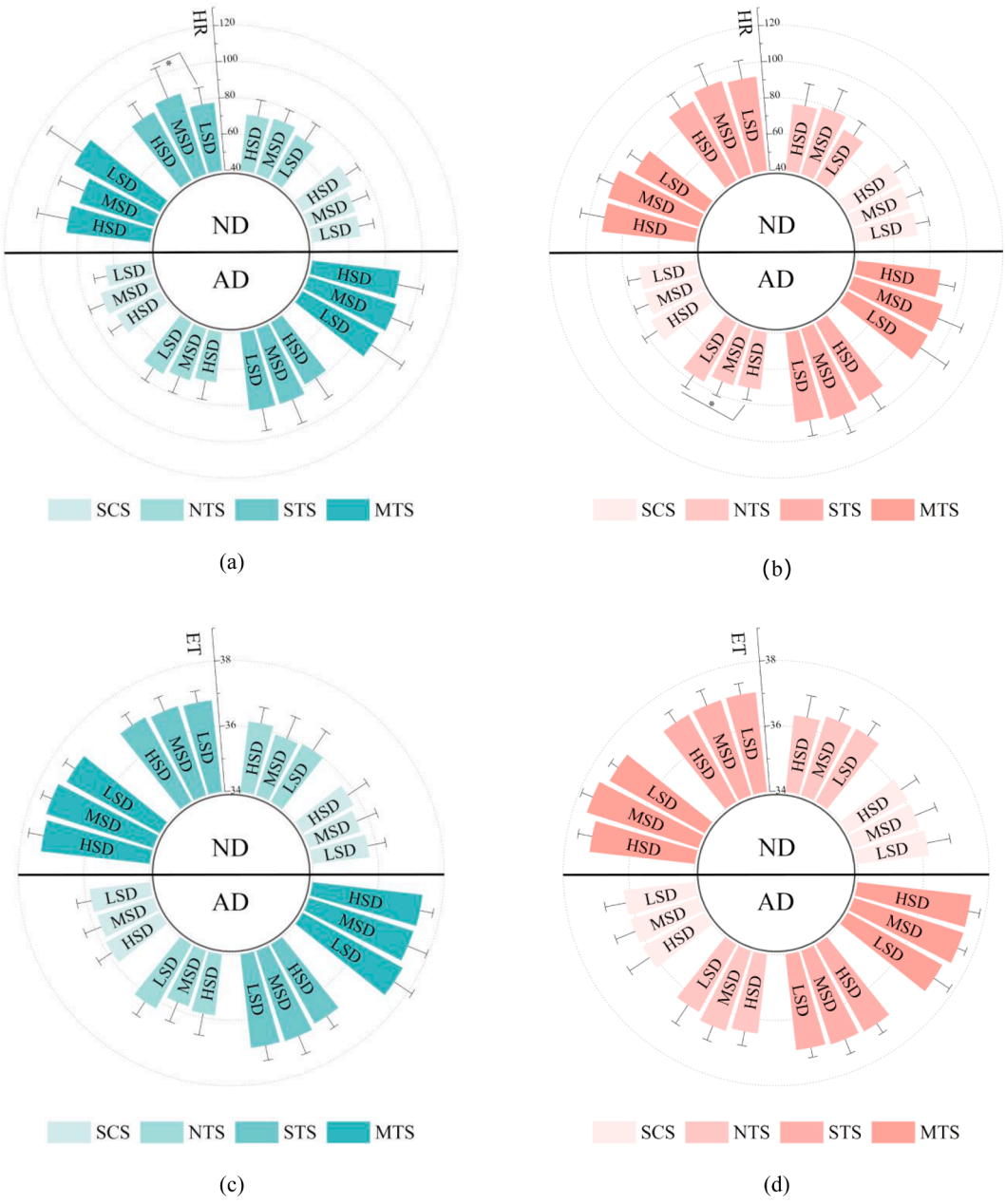

Fig. 9. Heart rate and eardrum temperature of male and female in different thermal stress and soundscape diversity across different sound source types: (a) male HR, (b) female HR, (c) male ET, (d) female ET.

3.3.3. EEG

Spectral decomposition (δ, θ, α, β) and topographical mapping were performed to characterize cortical activity under acoustic–thermal environments, with significance assessed via one-way ANOVA (*Appendix Table I – L*). Within natural environments, males exhibited elevated frontal δ power under MTS conditions, which may be consistent with heightened somnolence. LSD significantly attenuated this δ elevation relative to HSD ($p < 0.05$), suggesting potentially reduced drowsiness. Maximal α power emerged under NTS conditions in parieto-occipital regions, with LSD significantly elevating occipital α relative to MSD ($p < 0.05$) during MTS, indicating potentially enhanced relaxation (Fig. 11a).

Females demonstrated sexually dimorphic neurophysiological patterns. Under STS conditions, HSD significantly increased occipital δ power relative to MSD ($p < 0.05$), suggesting potential HSD-induced somnolence. Under MTS conditions, LSD significantly elevated occipital θ power relative to MSD ($p < 0.05$), which may be consistent with enhanced positive affect. Notably, α power exhibited a U-shaped trajectory with increasing diversity, indicating that MSD may have attenuated relaxation in females (Fig. 11b).

Within anthropogenic environments, sexually dimorphic neurophysiological responses emerged. Under MTS conditions, HSD significantly attenuated male δ power across frontal, parietal, and occipital regions relative to LSD ($p < 0.05$), suggesting potentially reduced somnolence. Conversely, LSD diminished frontal θ power versus MSD and occipital θ power versus HSD ($p < 0.05$), indicating potentially attenuated positive affect. The NTS conditions maximized parieto-occipital α power, with MSD exhibiting relatively elevated values. LSD significantly increased occipital β power versus HSD ($p < 0.05$) under NTS conditions, whereas HSD enhanced frontal, parietal, and occipital β power under MTS conditions, indicating potentially heightened alertness (Fig. 11c).

Table 2
Sex differences in ROS changes from Stage 1 to Stage 3 under nature-dominated soundscapes: effects of thermal stress and soundscape diversity.

Sex	Thermal stress	LSD			MSD			HSD		
		Stress	Attention	Cognition	Stress	Attention	Cognition	Stress	Attention	Cognition
Male	SCS	4.44 3.63 4.42	3.88 3.50 3.88	4.31 3.59 3.97	4.63 3.75 4.79	4.06 4.06 4.63	4.25 3.72 4.59	4.56 4.00 4.46	4.44 4.69 5.06	4.63 4.28 4.28
	NTS	5.47 3.78 4.47	4.20 4.20 4.07	5.00 4.07 4.17	5.02 4.12 4.19	4.07 4.29 3.57	4.64 4.00 3.68	4.35 3.98 3.88	3.85 3.90 3.70	4.23 3.98 3.90
	STS	4.55 4.49 4.25	3.52 3.94 3.48	4.58 4.40 4.24	4.38 3.90 4.14	3.58 3.92 3.62	4.35 3.81 4.12	4.97 4.42 4.88	3.86 3.86 4.09	4.80 4.25 4.45
	MTS	3.94 3.44 3.67	4.00 3.67 3.83	4.17 3.67 3.83	4.40 3.80 4.20	3.60 3.40 3.90	4.60 3.90 4.10	2.97 2.79 3.21	3.18 3.64 3.36	2.86 2.91 3.36
Female	SCS	4.65 3.86 4.53	3.88 3.47 3.41	4.56 3.85 4.59	4.67 3.85 4.71	3.79 3.89 3.57	4.38 3.95 4.05	4.84 4.12 4.84	4.09 4.04 4.17	4.61 4.24 4.50
	NTS	5.25 4.28 5.08	4.08 4.00 4.04	5.00 4.21 4.40	5.21 4.08 4.95	4.00 4.31 3.38	5.08 4.35 4.88	5.14 3.74 4.67	3.68 3.47 4.42	4.61 3.71 4.76
	STS	4.49 4.15 4.50	4.04 4.23 3.92	4.50 4.12 4.58	4.85 4.06 4.33	3.95 4.18 3.73	4.66 4.11 4.11	4.07 3.96 4.30	3.70 3.70 3.65	4.04 3.93 4.09
	MTS	4.63 3.58 4.13	4.00 3.50 3.50	4.63 3.75 4.06	3.87 3.28 3.79	3.46 3.38 3.83	3.46 3.54 3.77	3.47 3.23 3.43	3.40 3.20 3.50	3.50 3.55 3.55

Females demonstrated distinct spectral patterns. Under STS conditions, temporal δ power decreased linearly with acoustic diversity ($p < 0.05$), suggesting potential diversity-dependent somnolence reduction. θ power exhibited a U-shaped trajectory with minimal values at MSD, indicating that positive affect may have been attenuated at intermediate complexity. The MSD significantly elevated parietal and occipital α power relative to LSD ($p < 0.05$) during NTS. Under STS conditions, frontal and temporal β power increased with diversity ($p < 0.05$), suggesting potentially enhanced alertness (Fig. 11d).

3.4. Influencers of thermal- soundscape diversity perception, emotion and psychophysiological recovery

PLS-VIP analysis quantified the relative importance of soundscape diversity and micrometeorological parameters (T_g, V_a, T_a, *RH*, and T_{mrt}) as predictors of psychophysiological outcomes, including encompassing cardiovascular/thermal metrics (SBP, DBP, HR, ET) and psychological states (stress, cognition, attention, PA, NA, δ, θ, α, β), across sexes, acoustic contexts, and thermal stress gradients. This multivariate decomposition identified key determinants and elucidated sex-specific regulatory mechanisms underlying thermo-acoustic interactions.

Within natural environments, perceptual determinants exhibited thermally contingent, sexually dimorphic patterns. Under SCS conditions, male thermal perception (TSV/TCV) was predominantly driven by T_{mrt} and *RH*, whereas acoustic perception (SDV/SPV) showed elevated diversity importance; conversely, female perceptual indicators were predominantly diversity-mediated. Under NTS conditions, male thermal perception shifted to T_{mrt}/T_g dominance while acoustic perception retained diversity importance, whereas female indicators transitioned toward meteorological (T_{mrt}/T_g) dominance. Under STS conditions, male perception became primarily meteorologically regulated, while female thermal perception remained T_{mrt} -driven but acoustic perception was codetermined by diversity and *RH*. Under MTS conditions, male thermal perception was jointly influenced by *RH*, V_a, and diversity (with diversity maintaining high importance for acoustic perception), whereas female thermal and acoustic perception (except SDV driven by T_{mrt}) were predominantly diversity-mediated. Thus, soundscape diversity's predictive importance for male thermal perception intensified monotonically with escalating thermal stress, whereas female influence followed a U-shaped trajectory (initial attenuation followed by

Table 3
Sex differences in ROS changes from Stage 1 to Stage 3 under anthropogenic-dominant soundscapes: effects of thermal stress and soundscape diversity.

Sex	Thermal stress	LSD			MSD			HSD		
		Stress	Attention	Cognition	Stress	Attention	Cognition	Stress	Attention	Cognition
Male	SCS	4.65 3.94 3.67	3.06 3.94 3.69	4.34 4.00 3.88	4.81 3.91 4.57	3.94 3.83 4.00	4.50 3.83 4.31	4.69 3.84 4.39	3.94 4.18 3.88	4.47 4.00 4.29
	NTS	4.28 3.97 4.03	3.85 4.00 3.77	4.23 3.73 3.62	5.17 3.50 4.38	4.00 3.88 4.50	5.13 3.81 4.25	4.50 3.79 3.88	4.07 4.36 4.07	4.14 3.64 3.82
	STS	4.70 3.75 4.22	3.78 3.35 3.70	4.52 3.74 4.07	4.13 3.69 4.08	3.55 3.90 3.86	4.00 3.55 3.88	4.49 4.11 4.26	3.66 3.86 3.48	4.33 4.28 4.40
	MTS	2.70 2.58 2.76	3.18 3.18 3.00	2.64 2.45 2.86	3.24 2.76 2.67	3.43 3.14 2.86	3.36 3.00 3.00	3.17 2.96 3.38	3.00 3.00 3.13	3.13 3.06 3.44
Female	SCS	4.70 3.91 4.16	3.83 4.39 4.04	4.63 4.28 4.09	4.56 4.10 4.15	3.71 3.92 3.54	4.56 4.08 4.02	4.60 4.17 4.33	3.50 4.00 3.07	4.18 3.82 4.11
	NTS	5.30 4.48 3.97	4.30 4.35 4.26	4.98 4.37 4.13	5.42 4.28 4.52	4.43 4.35 3.83	5.15 3.98 4.15	5.06 4.10 4.48	4.07 4.37 3.59	4.70 4.33 4.11
	STS	4.17 3.37 3.91	3.58 3.54 3.81	3.83 3.38 3.52	4.40 4.10 4.17	4.00 3.86 3.83	4.22 3.97 3.95	4.74 4.15 4.58	3.96 4.04 4.50	4.60 4.35 4.21
	MTS	3.78 3.41 3.59	4.00 3.56 3.56	4.00 3.94 3.83	3.83 3.67 3.75	3.75 3.50 3.50	3.38 3.38 3.63	4.21 3.13 3.75	3.38 3.00 3.75	3.56 3.38 3.44

augmentation). For acoustic perception, diversity importance exhibited U-shaped patterns across thermal stress gradients for both sexes.

Physiological determinants exhibited thermally contingent, sexually dimorphic patterns. From cool-to-moderate thermal conditions (SCS, NTS and STS), male cardiovascular responsivity was codetermined by acoustic diversity and micrometeorological parameters; however, under MTS conditions, meteorological factors predominated. Conversely, female physiological indicators were initially meteorologically dominated (SCS) but subsequently demonstrated joint diversity-meteorological regulation (NTS, STS and MTS). Thus, soundscape diversity's predictive importance for male physiological responses monotonically attenuated with escalating thermal stress, whereas female responses exhibited progressive intensification of acoustic influence.

Psychological determinants exhibited thermally contingent, sexually dimorphic regulatory patterns. Male affective and cognitive responses were predominantly diversity-driven under SCS conditions, transitioning to joint diversity-meteorological codetermination across NTS to MTS conditions. Conversely, female responses were characterized by consistent joint regulation across SCS to MTS conditions, yet exclusively meteorologically dominated under NTS conditions. Consequently, male psychological indicators demonstrated heightened sensitivity to acoustic complexity with marked stage-dependent volatility, whereas female responses exhibited a biphasic trajectory in the influence of diversity, with initial attenuation followed by augmentation, under escalating thermal stress (Fig. 12a, b).

Within anthropogenic environments, predictor importance exhibited thermally contingent sexual dimorphism. Perceptually, males demonstrated persistent meteorological dominance for thermal sensation/comfort across thermal gradients, with acoustic diversity primarily governing soundscape discrimination (SDV) under cool-to-moderate stress but attenuating under MTS conditions. Conversely, females maintained acoustic dominance for soundscape appraisal (SDV, SPV) across conditions, while thermal perception showed non-monotonic diversity modulation, with initial intensification followed by attenuation under escalating stress, against sustained meteorological influence.

Physiologically, male cardiovascular and thermal responses consistently exhibited joint diversity-meteorological codetermination across all thermal stages. Female indicators, however, transitioned from initial codetermined regulation (SCS) to meteorological dominance (NTS and STS), reverting to joint regulation under MTS conditions, resulting in

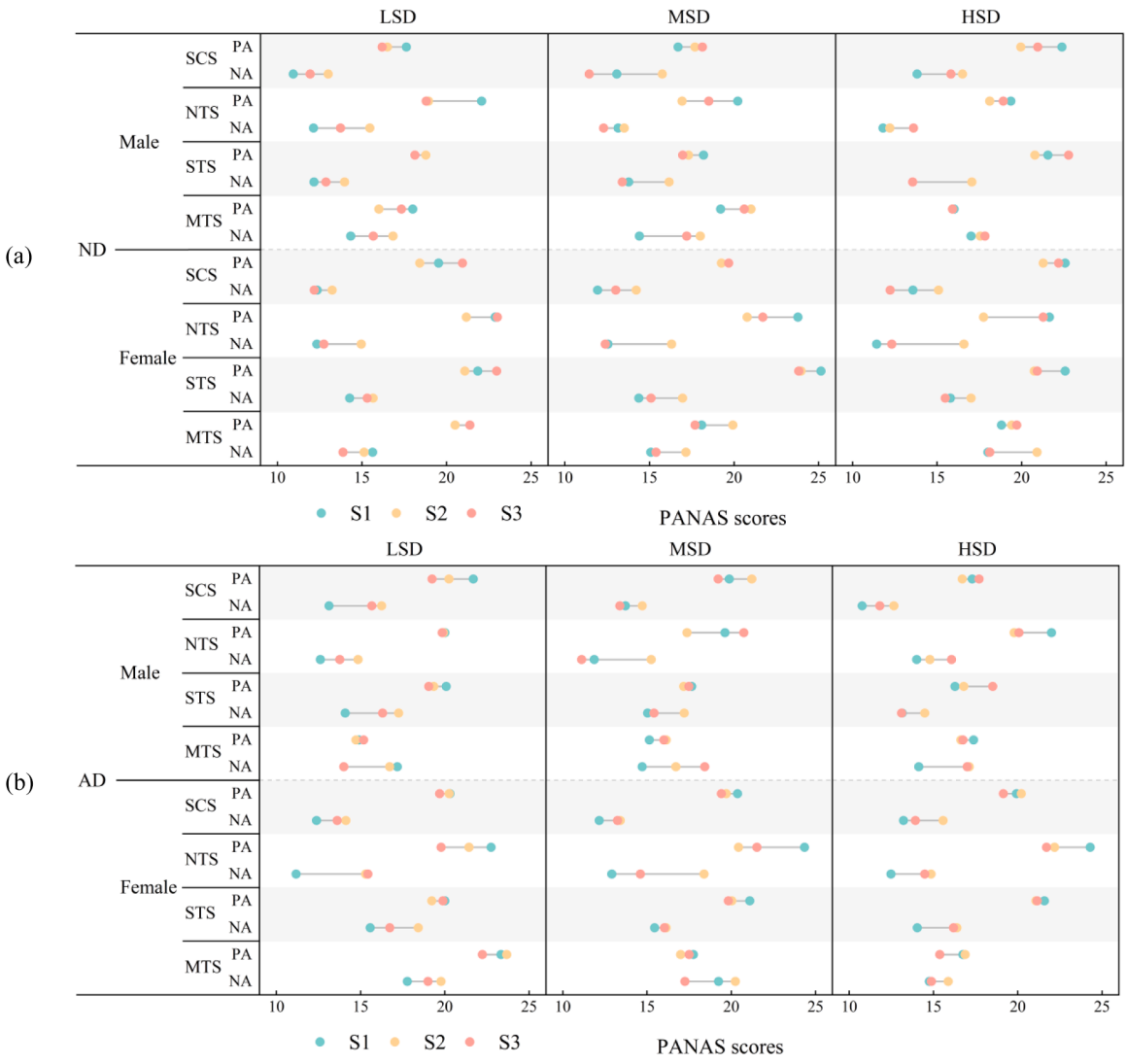

Fig. 10. Scores for PA and NA: (a) ND, (b) AD.

progressive intensification of acoustic influence with thermal escalation.

Psychologically, male affective and cognitive states demonstrated consistent joint regulation by acoustic complexity and micrometeorological parameters across all thermal conditions. Female psychological responses shifted from exclusive meteorological dominance (SCS and NTS) to joint diversity-meteorological codetermination (STS and MTS). Consequently, male psychological indicators maintained stable acoustic sensitivity with marked stage-dependent volatility, whereas female diversity influence monotonically strengthened with escalating thermal stress (Fig. 12c, d).

4. Discussion

4.1. Thermal and soundscape perceptions

Our findings demonstrate that significant thermo-modulatory effects of acoustic diversity were confined to natural contexts, consistent with established restorative properties of natural soundscapes [10,64]. Anthropogenic acoustic environments, by contrast, elicit negative appraisals that attenuate perceived environmental quality [65], potentially masking the salutogenic effects of acoustic diversity and thereby attenuating its thermoregulatory influence.

Within natural contexts under slight cold stress, HSD significantly attenuated female subjective cold sensation while exerting minimal male effect. This sexual dimorphism may reflect enhanced female auditory sensitivity and attentional engagement with environmental acoustic detail [66], whereby enriched natural soundscapes capture attentional resources and divert cognitive focus from thermal discomfort. Conversely, under STS conditions, LSD significantly improved male thermal comfort. This pattern likely reflects sex-specific cognitive load management: elevated temperatures impair male executive function and increase cognitive demands [67], and LSD may provide a lower-load acoustic environment that mitigates cognitive stress and enhances thermal comfort.

Sex and thermal stress interactively govern acoustic diversity perception. Individual attributes substantially modulate soundscape appraisal, with sexual dimorphism particularly salient [68,69]. Thermal conditions further constrain perceptual processing through affective and attentional mechanisms, thereby modulating acoustic evaluation [32,70].

Within anthropogenic environments, LSD significantly attenuated male pleasantness (SPV) under slight cold stress, whereas female responsivity remained muted. This sexual dimorphism likely reflects inadequate acoustic masking: monotonous LSD insufficiently obscures negative sources, amplifying perceived noise exposure persistence and degrading male hedonic response [71]. Conversely, females demonstrate heightened susceptibility to acoustic intensity and noise-related attributes, potentially attenuating the regulatory efficacy of diversity

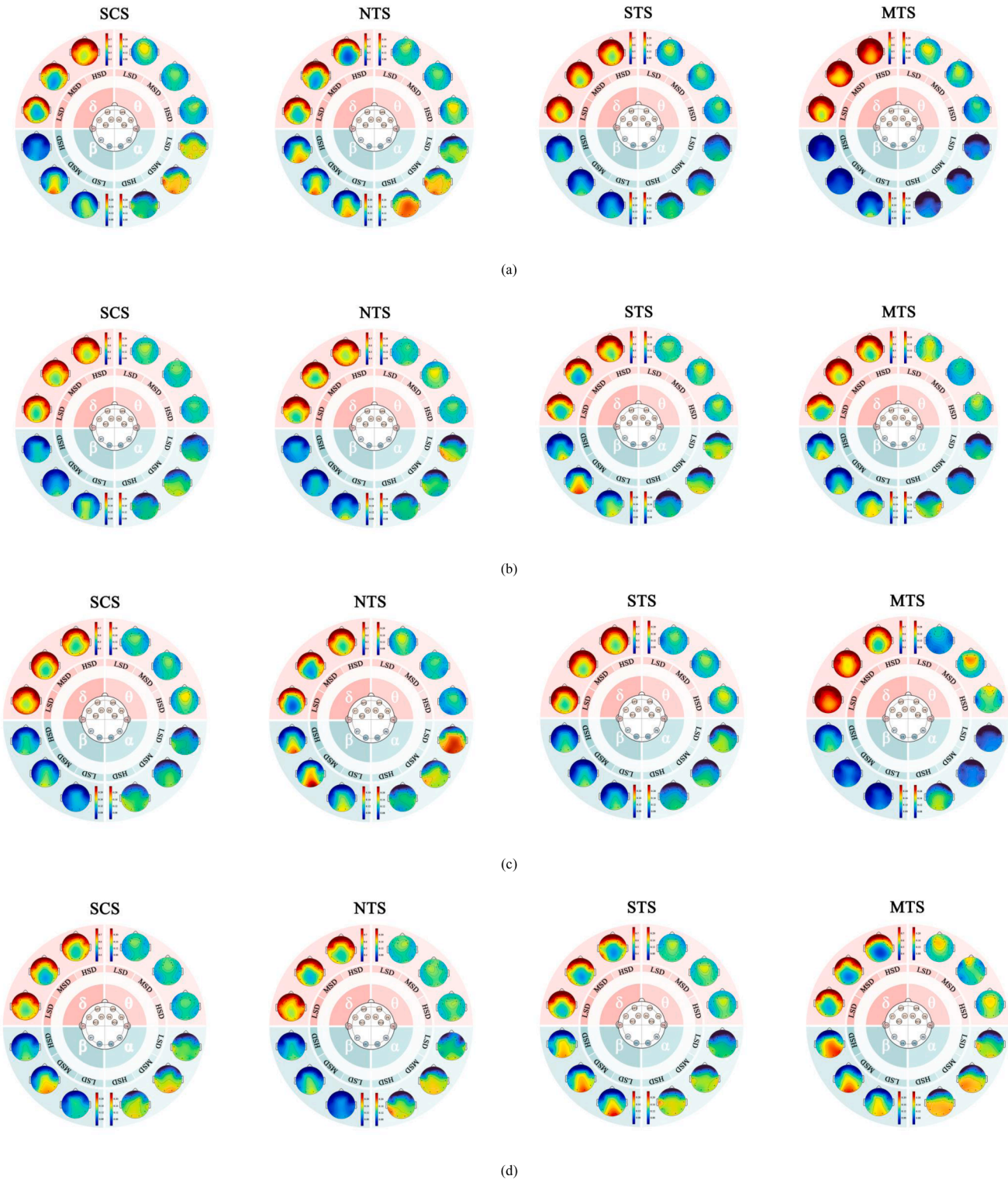

Fig. 11. Topographical distributions and developments of δ, θ, α and β wave normalized PSD: (a) ND-male, (b) ND-female, (c) AD-male, (d) AD-female.

manipulation on their perceptual evaluations.

4.2. Physiological responses

Sexual dimorphism characterized cardiovascular responsivity to acoustic diversity across dominant source contexts. Within natural environments, diversity significantly modulated male hemodynamic and chronotropic parameters (SBP, DBP, HR) under STS conditions, contrasting with female non-responsivity. Within anthropogenic environments, diversity effects were markedly attenuated, with females exhibiting marginal chronotropic responsivity solely under NTS conditions. These findings indicate heightened male autonomic sensitivity to acoustic structural complexity, who demonstrated pronounced male heart rate deceleration following aversive acoustic exposure [72]. However, these results contrast with Bai and Zhang, who reported negligible sexual dimorphism in psychophysiological responses to forest

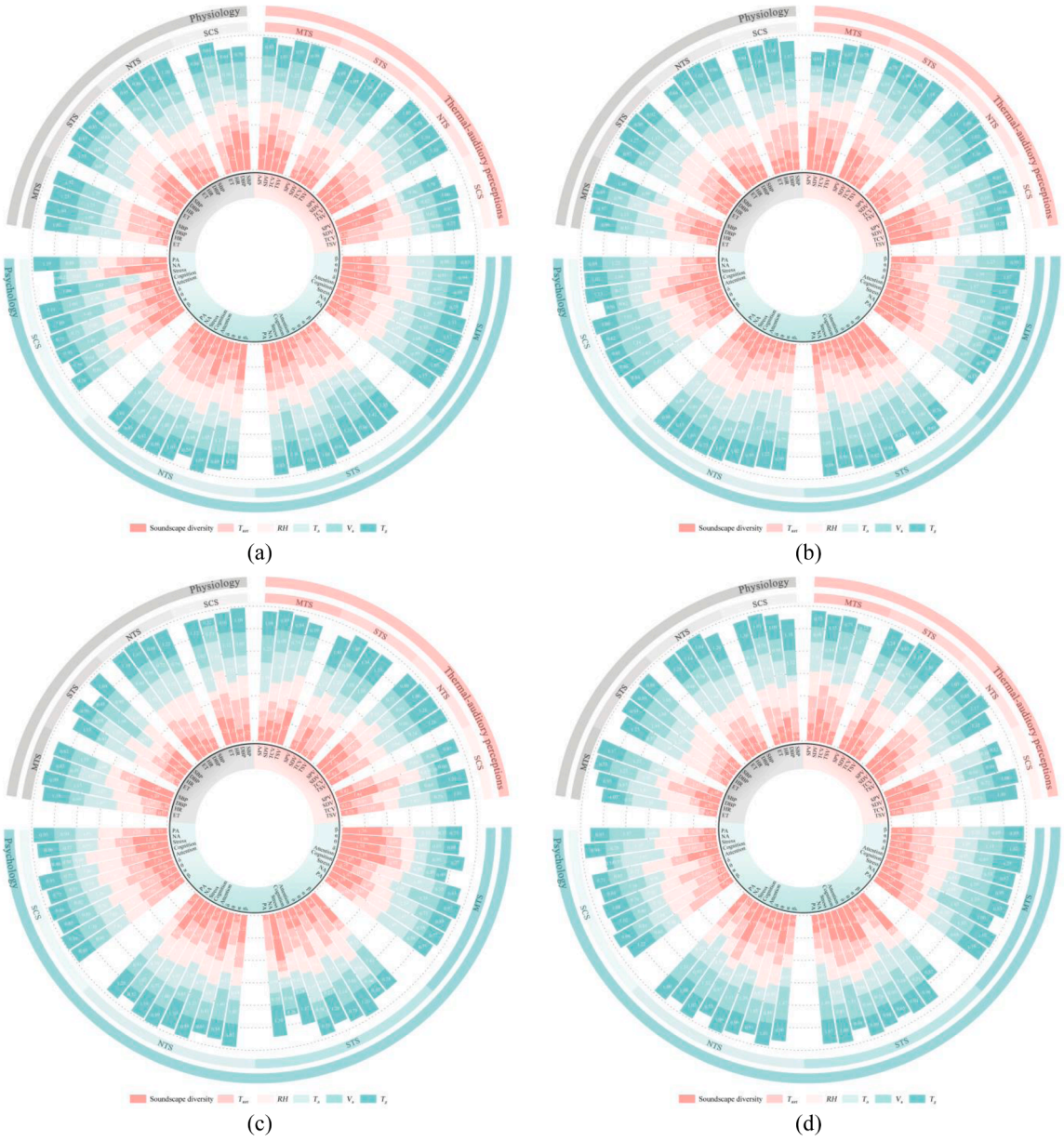

Fig. 12. Dominant influencers of thermal- soundscape diversity perception, emotion and psychophysiological recovery: (a) ND-male, (b) ND-female, (c) AD-male, (d) AD-female.

park soundscapes under controlled indoor conditions [34]. This discrepancy likely reflects the intensified thermal complexity of outdoor environments, which amplifies individual sensitivity to acoustic variation and unmasks latent sexual dimorphism in autonomic regulation. The established salutogenic efficacy of natural environments may further potentiate the detection of sex-specific cardiovascular modulation under thermal challenge [5,64,73].

Under STS conditions, male cardiovascular responses to natural diversity exhibited parameter-specific optima: bradycardia under LSD (parasympathetic predominance) [74], reduced systolic pressure under MSD (optimal arousal) [75], and attenuated diastolic pressure under HSD (engagement-mediated vasodilation) [64,76]. These divergent chronotropic, inotropic, and vascular responses indicate that acoustic complexity modulates autonomic function through distinct physiological pathways. Notably, these effects were thermally contingent, manifesting primarily under STS conditions, underscoring that thermal load modulates acoustic sensitivity [77,78].

4.3. Psychological effects

Our findings suggest that soundscape diversity may contribute to psychological restoration. Within natural environments, thermal escalation potentiated male cortical responsivity to diversity, whereas females exhibited increasing θ responsivity. This aligns with established restorative potentiation under environmental challenge [14,79].

Within anthropogenic environments, male spectral responsivity (δ, θ, β) intensified monotonically with thermal stress, whereas females demonstrated restricted diversity sensitivity confined to NTS and STS conditions. This sexual dimorphism may reflect differential attentional resource allocation under environmental load [80]. Moderate stress may facilitate external information processing; however, escalating thermal challenge may prioritize physiological coping, thereby attenuating environmental detail perception [75]. Thermal stress alters EEG dynamics and recruits cognitive resources [81], with combined acoustic–thermal demands potentially degrading cognitive performance [82]. Consequently, females may retain diversity sensitivity under mild

thermal conditions, with responsivity attenuating at thermal extremes due to attentional resource competition.

Within natural environments under MTS conditions, sexual dimorphism characterized neurophysiological restoration. Males exhibited attenuated frontal δ power coupled with enhanced occipital α activity, concomitant with improved attention, stress recovery, and affective valence, suggesting potentially heightened relaxation and restoration. Females demonstrated elevated θ power with parallel psychometric improvements, which may be consistent with enhanced positive affect and well-being [17]. These outdoor findings diverge from Bai et al., who reported superior restoration from composite versus single-source natural sounds indoors [34]. This discrepancy may reflect the elevated perceptual complexity of outdoor thermally challenging contexts, wherein HSD (incorporating anthropogenic elements) may impose excessive cognitive load, attenuating the restorative efficacy observed in controlled indoor settings.

Within anthropogenic environments, males under MTS conditions exhibited HSD-mediated attenuation of δ power across frontoparietal and occipital regions, coupled with enhanced θ and β activity and improved psychometric indices. This pattern suggests that acoustic complexity may facilitate attentional engagement, potentially mitigating cognitive resource depletion induced by thermal challenge [83]. Conversely, females demonstrated context-dependent optima. Under NTS conditions, MSD maximized parieto-occipital α power with concomitant stress reduction, potentially indicating balanced sensory stimulation avoiding both monotony and excessive load [84]. Under STS conditions, HSD reduced δ while enhancing β activity, which may have facilitated stress, attention, and affective recovery, potentially reflecting complexity-mediated replenishment of depleted attentional resources under environmental challenge [85].

4.4. Limitations and future directions

This study examined acoustic diversity effects from a sex-differentiated perspective. However, the sample comprised university students with homogeneous demographic characteristics, potentially constraining external validity. Subsequent research should stratify participants across age cohorts, occupations, and cultural contexts, incorporating anthropometric and metabolic indices (BMI and metabolic rate) to elucidate population-specific psychophysiological responsivity. Second, acoustic stimuli were experimentally synthesized using controlled diversity indices, potentially limiting ecological validity relative to in situ acoustic heterogeneity. Future investigations should explore extended diversity gradients using ecologically authentic soundscapes or immersive virtual environments. Finally, the acute exposure protocol precludes assessment of chronic adaptation dynamics. Longitudinal or long-term field monitoring designs are warranted to characterize cumulative exposure effects, habituation mechanisms, and potential interactions with olfactory and visual environmental complexity in multisensory urban contexts.

5. Conclusions

This study investigated sexual dimorphism in thermo-acoustic responsivity across three microclimatic contexts (open, partially shaded, shaded) within an urban green space in Xi'an, China. Thermal stress was operationalized via four PET categories. Eight hundred sixty-four university students (sex ratio 1:1.2) underwent orthogonal exposure to six acoustic conditions: LSD, MSD and HSD within natural and anthropogenic-dominant environments. Multimodal assessment integrated micrometeorological monitoring, psychometric evaluation, cardiovascular and thermal physiology, and electroencephalography to characterize sex-differentiated psychophysiological responsivity. Principal findings indicate:

1) Sexual dimorphism characterized thermo-perceptual responsivity predominantly within natural environments: males demonstrated significant TCV modulation under STS conditions, SDV sensitivity under SCS and MTS conditions, and SPV responsivity under SCS conditions ($p < 0.05$), whereas females exhibited exclusive TSV sensitivity under SCS conditions ($p < 0.05$). Within anthropogenic environments, diversity exerted no significant thermo-perceptual effects, though SDV responsivity emerged under SCS in both sexes ($p < 0.05$) and SPV under NTS ($p < 0.05$), with females demonstrating heightened SDV sensitivity ($p < 0.001$).
2) Cardiovascular responsivity demonstrated sexual dimorphism and context-dependency, whereas ET responses were non-significant. Within natural environments under STS conditions, males exhibited significant hemodynamic attenuation: SBP decreased 6.08 mmHg (MSD vs. LSD) and DBP decreased 6.61 mmHg (HSD vs. LSD; $p < 0.05$). Females demonstrated significant chronotropic reduction exclusively under NTS anthropogenic conditions (HR decreased 5.32 bpm, HSD vs. LSD; $p < 0.05$).
3) Under MTS within natural environments, neuropsychological restoration exhibited sex-convergent efficacy with divergent neural signatures. LSD significantly attenuated male frontal δ power while enhancing occipital α power ($p < 0.05$), concomitant with improved attention, stress recovery, and affective valence. Females exhibited elevated θ power ($p < 0.05$) with parallel improvements in stress recovery, cognition, and affective state. Within anthropogenic environments, diversity effects on psychological states were more heterogeneous.
4) Thermal stress gradients elicited sexually dimorphic, domain-specific diversity effects. Within biophonic environments, male thermal perception strengthened monotonically with escalating stress, whereas females exhibited biphasic (attenuation-recovery) trajectories; soundscape perception followed biphasic patterns across sexes. Physiologically, male responsivity attenuated while female responsivity intensified; psychologically, males maintained heightened sensitivity, whereas females demonstrated biphasic trajectories. Within anthropogenic environments, male thermal perception strengthened monotonically, whereas females exhibited inverted-U patterns; soundscape perception demonstrated inverted-U trajectories in males and monotonic attenuation in females. Males exhibited greater overall psychophysiological sensitivity; females demonstrated U-shaped physiological responsivity and progressively intensifying psychological responsivity.

CRediT authorship contribution statement

Zihan Tang: Writing – original draft, Investigation, Data curation. **Wenxia Huang:** Investigation, Data curation. **Miaosen Zhang:** Investigation, Data curation. **Pei Deng:** Investigation, Data curation. **Bo Hong:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in

the online version, at doi:10.1016/j.buildenv.2026.114742.

Data availability

Data will be made available on request.

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